Global Fuel Prices and Taxation Analysis (2020–2026)

Step-by-Step Project Execution & Student Report

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STEP 1 PROBLEM FORMULATION & PROJECT SCOPE

1.1 Background & Problem Statement

Energy commodities (Petrol, Diesel, LPG) represent fundamental cost drivers across global supply chains and national economies. However, pricing structures are heavily distorted by regional tax regimes, carbon levies, and national subsidy programs. Because data regarding retail prices, tax burdens, and raw crude benchmarks is scattered across disparate international sources, decision-makers struggle to gain a unified analytical perspective.

Core Challenge: Without a centralized analytical system, policy advisors cannot evaluate fiscal tax impacts, freight directors struggle to manage fuel price volatility, and energy economists lack clear pass-through models linking Brent crude rates to pump prices.

1.2 Project Objectives

- Construct a unified data pipeline integrating 27,468 international fuel price records across 84 countries and 7 global regions (2020–2026).
- Develop interactive Tableau dashboards providing customized analytics for Policy Consultants, Logistics Managers, and Energy Economists.
- Measure the mathematical correlation between Brent crude spot prices and retail consumer rates to identify price transmission lag times.

1.3 Key Stakeholder Personas

Persona	Role	Core Need / Analytical Focus
Dr. Sarah Mitchell	Policy Consultant	Evaluate tax percentage share per country, compare subsidy burdens, and support national carbon tax reform.
Marcus Thompson	Supply Chain Director	Track multi-year diesel price volatility, locate low-cost freight refueling corridors, and forecast 12-month trends.
Prof. Elena Rodriguez	Energy Economist	Model Brent crude spot price pass-through dynamics and evaluate fuel affordability across country income tiers.

STEP 2 DATA COLLECTION & PREPROCESSING

2.1 Dataset Structure & Attributes

The underlying dataset consists of **27,468 records** spanning 2020 to 2026. The structured CSV dataset includes the following key variables:

- `Country` & `Region`: ISO geography codes covering 84 nations across 7 world regions.
- `Fuel Type`: Commodity categories (Petrol, Diesel, LPG).
- `Pump Price (USD)`: Final retail fuel price per liter normalized to US Dollars.
- `Tax Percentage (%)`: Government tax component share relative to retail price.
- `Subsidy Indicator & Amount`: Binary flag and net subsidy expenditure ($ USD) per liter.
- `Brent Crude Spot Price ($/bbl)`: Benchmark international raw crude oil spot rates.
- `Income Level Tier`: World Bank economic classification (High, Upper-Middle, Lower-Middle, Low).

2.2 Data Cleaning & Transformation Workflow

Prior to visual development in Tableau, raw records underwent rigorous data cleaning and transformation procedures:

- **Currency Normalization:** Converted local currency pump rates to standardized USD per liter values using historical annual exchange averages.
- **Null & Outlier Management:** Removed duplicate record logs and imputed missing regional tax percentages using country-level historic medians.
- **Geographic Role Assignment:** Configured `Country` and `Region` fields in Tableau as geographical dimensions to enable automated map rendering.

2.3 Analytical Calculated Fields Created in Tableau

```
1. [Net Fuel Price] = [Pump Price (USD)] * (1 - ([Tax Percentage] / 100)) + [Subsidy Amount]
2. [Tax Amount ($)] = [Pump Price (USD)] * ([Tax Percentage] / 100)
3. [Volatility Index] = (WINDOW_STDEV(AVG([Pump Price (USD)]))) / WINDOW_AVG(AVG([Pump Price (USD)]))
4. [Crude Correlation] = CORR(AVG([Brent Crude Spot Price]), AVG([Pump Price (USD)]))
```

STEP 3 VISUALIZATION DEVELOPMENT & DASHBOARD DESIGN

3.1 Business Questions & Worksheet Construction

Twelve distinct worksheets were created in Tableau Desktop to address specific business questions framed around project personas:

- **Worksheet 1 (Tax Heatmap):** Geographical filled map rendering tax percentage intensity across 84 countries.
- **Worksheet 2 (Subsidy Packed Bubbles):** Proportional bubble chart illustrating net national subsidy volumes.
- **Worksheet 3 (Diesel Box Plot):** Distribution chart plotting diesel price spreads across international freight corridors.
- **Worksheet 4 (Crude Pass-Through Model):** Dual-axis line chart overlaying Brent crude spot prices with retail petrol pump rates.
- **Worksheet 5 (Income Tier Scatter Plot):** Scatter plot comparing GDP per capita against fuel price volatility.

3.2 Dashboard Architecture & User Interface

Worksheets were integrated into two interactive, scenario-driven dashboards connected by a unified Tableau Story:

Dashboard View	Included Visualizations	Interactive Controls & Features
Dashboard 1: Policy & Tax Structure View	Top KPI Summary Tiles, Geographic Tax Heatmap, Subsidy Packed Bubbles, Tax % vs. Price Scatter Plot.	Global Quick-Filters for Region and Fuel Type, dynamic tooltips displaying net tax amounts.
Dashboard 2: Logistics & Research View	2020–2026 Price Trend Lines, Brent Crude Pass-Through Model, Income Tier Cross-Tabs, Volatility Forecasts.	Parameter sliders for crude shock scenarios, country income tier highlighting.

STEP 4 ANALYSIS & KEY RESULTS

4.1 Summary Metric Baselines

Analysis of the 2020–2026 dataset yields the following baseline macro aggregates across the 84 studied countries:

- **Global Average Retail Petrol Price:** $1.42 per liter.
- **Global Average Tax Share %:** 41.8% of retail pump price.
- **Total International Subsidy Expenditure:** $95.4 Billion USD.
- **Brent Crude Correlation Coefficient:** r = 0.84 (strong positive relationship).

4.2 Detailed Findings by Analytical Domain

1. Fiscal Policy & Taxation Dynamics

European nations exhibit the highest fiscal tax burdens globally, with tax percentages averaging **58% to 62%** of retail pump prices. In contrast, North American tax shares remain moderate at 18% to 25%. Middle Eastern and North African economies utilize heavy net subsidies ($95.4B combined), keeping retail consumer prices artificially low.

2. Supply Chain & Freight Corridor Volatility

Diesel price distribution analysis reveals significant cost variances across key freight transit corridors. Freight routes in South Asia and Latin America demonstrate high 12-month price volatility (Volatility Index = 0.34), creating financial forecasting risks for logistics operators.

3. Econometric Brent Crude Pass-Through

Dual-axis correlation modeling confirms that raw Brent crude spot price shocks transmit to consumer pump rates with an estimated **3-to-4 week lag**. The strong correlation coefficient (r = 0.84) indicates high pass-through efficiency in non-subsidized markets.

STEP 5 SOLUTION TESTING & TECHNICAL EVALUATION

5.1 Workbook Performance Testing

The completed Tableau workbook (.twbx) underwent rigorous performance optimization checks via Tableau's native Workbook Auditor:

Test Parameter	Test Execution Details	Result
Query Render Speed	Extracted 27,468 rows into a packaged Hyper extract (.hyper). Average query response time < 0.8 seconds.	**PASSED**
Filter Responsiveness	Global quick-filters tested across all 12 worksheets simultaneously without UI lag.	**PASSED**
Rule Validation	17 out of 17 Tableau performance best practices rules passed without critical warnings.	**PASSED**

5.2 Project Advantages & Limitations

Advantages:

- Consolidates fragmented global energy data into one interactive, visual BI environment.
- Tailors dedicated views for three key professional personas (Policy, Freight, Research).
- Facilitates seamless online sharing and executive briefing via Tableau Public.

Limitations:

- Relies on static historical batch extracts (2020–2026) rather than live real-time exchange API feeds.
- High multi-country density requires filter selection for optimal small-screen viewing.

STEP 6 CONCLUSION & DELIVERABLES

6.1 Final Project Summary

This project successfully demonstrates the step-by-step transformation of raw, complex global energy commodity data into an interactive Business Intelligence solution. By completing problem definition, data preprocessing, analytical measure modeling, and dashboard construction in Tableau Desktop, the project delivers clear visual clarity on international fuel tax structures, logistics volatility, and crude pass-through dynamics.

6.2 Published Project Repositories

Live Interactive Tableau Public Story Link:
https://public.tableau.com/views/GlobalFuelPricesAndTaxationAnalysis2020-2026/Story1?:language=en-US&:display_count=n&:origin=viz_share_link

GitHub Code & Documentation Repository:
https://github.com/DivyanshiJain/GlobalFuelPricesAndTaxationAnalysis

Step-by-Step Project Sign-Off:

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